

PRACTICAL

TASK-2 :

1. Execution

```
bindhu@LAPTOP-NQPCRIKO: ~$ nano task2.c
bindhu@LAPTOP-NQPCRIKO: ~$ gcc task2.c
bindhu@LAPTOP-NQPCRIKO: ~$ ./a.out
^C
bindhu@LAPTOP-NQPCRIKO: ~$ ls -l b2 c2
ls: cannot access 'b2': No such file or directory
-rw-r--r-- 1 bindhu bindhu 89 Aug  3 08:37 c2
bindhu@LAPTOP-NQPCRIKO: ~$ echo "Hello World" > b2
bindhu@LAPTOP-NQPCRIKO: ~$ ./a.out
copied contents of b2 to c2
bindhu@LAPTOP-NQPCRIKO: ~$ cat c2
Hello World
program related to system calls
It is about open read and write system calls
bindhu@LAPTOP-NQPCRIKO: ~$ echo "Hello World" > sample.txt
bindhu@LAPTOP-NQPCRIKO: ~$ cat sample.txt
Hello World
bindhu@LAPTOP-NQPCRIKO: ~$ strace cat sample.txt
Command 'strace' not found, but can be installed with:
sudo apt install strace
bindhu@LAPTOP-NQPCRIKO: ~$ sudo apt install strace
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
Installing:
  strace

Installing dependencies:
  libunwind8

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 25
  Download size: 737 kB
  Space needed: 2660 kB / 1024 GB available

Continue? [Y/n] Y
Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libunwind8 amd64 1.8.3-0ubuntu1 [59.6 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 strace amd64 6.19+ds-0ubuntu5 [678 kB]
Fetched 737 kB in 1s (923 kB/s)
```

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bindhu@LAPTOP-NQPCRIKO: ~$ strace cat sample.txt
execve("/usr/bin/cat", ["cat", "sample.txt"], 0x7ffe7a38df28 /* 25 vars */) = 0
brk(NULL) = 0x55e655459000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x78ee6f052000
access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=20087, ...}) = 0
mmap(NULL, 20087, PROT_READ, MAP_PRIVATE, 3, 0) = 0x78ee6f04d000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0\0\0...", 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=199120, ...}) = 0
mmap(NULL, 206288, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x78ee6f01a000
mmap(0x78ee6f021000, 135168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7000) = 0x78ee6f021000
mmap(0x78ee6f042000, 28672, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x78ee6f042000
mmap(0x78ee6f049000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2f000) = 0x78ee6f049000
mmap(0x78ee6f04b000, 5584, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x78ee6f04b000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc_s.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0\0\0...", 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=187120, ...}) = 0
mmap(NULL, 185256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x78ee6efec000
mmap(0x78ee6eff0000, 147456, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x4000) = 0x78ee6eff0000
mmap(0x78ee6f014000, 16384, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x78ee6f014000
mmap(0x78ee6f018000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2c000) = 0x78ee6f018000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libm.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0\0\0...", 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=1198376, ...}) = 0
mmap(NULL, 1200152, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x78ee6e6c0000
mmap(0x78ee6e6d8000, 655360, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x12000) = 0x78ee6e6d8000
mmap(0x78ee6ef78000, 466944, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xb2000) = 0x78ee6ef78000
mmap(0x78ee6efea000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x123000) = 0x78ee6efea000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0\0\0...", 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0...", 840, 64) = 840

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bindhu@LAPTOP-NQPCRIKO: ~$ strace cat sample.txt
openat(AT_FDCWD, "/usr/share/coreutils/locales/uucore/en-US.ftl", O_RDONLY|O_CLOEXEC) = 3
statx(3, "", AT_STATX_SYNC_AS_STAT|AT_EMPTY_PATH, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFREG|0644, stx_size=4080, ...}) = 0
read(3, "# Common strings shared across a...", 4080) = 4080
read(3, "", 32) = 0
close(3) = 0
statx(AT_FDCWD, "/usr/share/coreutils/locales/cat", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/cat/en-US.ftl", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
rt_sigaction(SIGPIPE, {sa_handler=SIG_DFL, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x78ee6ec45cb0}, {sa_handler=SIG_IGN, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x78ee6ec45cb0}, 0) = 0
rt_sigaction(SIGSEGV, {sa_handler=SIG_DFL, sa_mask=[RTMIN RT_1], sa_flags=SA_RESTORER, sa_restorer=0x78ee6ec45cb0}, {sa_handler=0x55e64d8d60c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x78ee6ec45cb0}, 0) = 0
rt_sigaction(SIGBUS, {sa_handler=SIG_DFL, sa_mask=[RTMIN RT_1], sa_flags=SA_RESTORER, sa_restorer=0x78ee6ec45cb0}, {sa_handler=0x55e64d8d60c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x78ee6ec45cb0}, 0) = 0
ioctl(1, TCGETS2, {c_iflag=ICRNL|IXON, c_oflag=NL0|CR0|TAB0|BS0|VT0|FF0|POST|ONLCR, c_cflag=B38400|CS8|CREAD, c_lflag=ISIG|ICANON|ECHO|ECHOE|ECHOK|TEXTEN|ECHOCTL|ECHOKE, ...}) = 0
brk(0x55e65549b000) = 0x55e65549b000
statx(AT_FDCWD, "sample.txt", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFREG|0644, stx_size=12, ...}) = 0
openat(AT_FDCWD, "sample.txt", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=12, ...}) = 0
fstat(1, {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0x2), ...}) = 0
splice(3, NULL, 1, NULL, 1048576, 0) = -1 EINVAL (Invalid argument)
pipe([4, 5]) = 0
fcntl(4, F_SETPIPE_SZ, 1048576) = 1048576
splice(3, NULL, 5, NULL, 1048576, 0) = 12
splice(4, NULL, 1, NULL, 12, 0>Hello World) = 12
splice(3, NULL, 5, NULL, 1048576, 0) = 0
close(5) = 0
close(4) = 0
close(3) = 0
sigaltstack({ss_sp=NULL, ss_flags=SS_DISABLE, ss_size=8192}, NULL) = 0
munmap(0x78ee6f04f000, 12288) = 0
exit_group(0) = ?
+++ exited with 0 +++
bindhu@LAPTOP-NQPCRIKO:~$

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